THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a party 15 people greet each other once with a hug. Number of hugs: 1

(a) 106 **(b)** 90 **(c)** 105 **(d)** 120

2. The sum $1 + 2 + 3 + \cdots + n$ equals 703. Then n is: 1

(a) 37 **(b)** 38 **(c)** 39 **(d)** 36

3. The sum of the first 30 odd numbers is 900. The sum of the odd numbers from 21 to 59 is: 1

(a) 799 **(b)** 800 **(c)** 810 **(d)** 801

4. The difference between the squares of the 20th and 19th numbers is: 1

(a) 40 **(b)** 39 **(c)** 38 **(d)** 41

5. The first term of a pattern is 7 and the 14th term is 150. The common difference is: 1

(a) 10 **(b)** 11 **(c)** 12 **(d)** 13

6. A pattern adds 1, then 2, then 3, ... Starting at 10, the term after 12 additions is: 1

(a) 88 **(b)** 100 **(c)** 87 **(d)** 89

7. The sum $1^3 + 2^3 + \cdots + 9^3$ equals: 1

(a) 2024 **(b)** 2026 **(c)** 2070 **(d)** 2025

8. The 41th triangular number minus the 40th triangular number is: 1

(a) 43 **(b)** 42 **(c)** 40 **(d)** 41

9. In a party 18 people greet each other once with a hug. Number of hugs: 1

(a) 153 **(b)** 171 **(c)** 135 **(d)** 154

10. The sum $1 + 2 + 3 + \cdots + n$ equals 990. Then n is: 1

(a) 45 **(b)** 44 **(c)** 46 **(d)** 43

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. 14 cricket teams play a league in which every team plays every other team once. Find the number of matches. 2

12. The sum of the first n odd numbers is 961. Find n and the 31th odd number. 2

13. Find the sum of odd numbers from 19 to 47. 2

14. 16 cricket teams play a league in which every team plays every other team once. Find the number of matches. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a pattern the 1st term is 14, the 2nd is 23, and so on by adding 9. Which term of the pattern equals 365? Find also the term just before it. 3

16. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 39 and use it to find 20^2. Then evaluate the sum from 41 to 49. 3

17. A tower is built with rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 6, and in the first 6 rows together? 3

OR

In a pattern the 1st term is 23, the 2nd is 30, and so on by adding 7. Which term of the pattern equals 331? Find also the term just before it. 3

18. Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 49 and use it to find 25^2. Then evaluate the sum from 51 to 61. 3

OR

A tower is built with rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 7, and in the first 7 rows together? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Look at the pattern: the 8th figure has 46 dots. How many dots are in figure 15? If a figure has 118 dots, which figure is it? Show that the dots form a pattern with common difference 6. 5

OR

Two sequences: A is $5, 9, 13, 17, \ldots$ and B is $20, 22, 24, 26, \ldots$. Find the 12th term of each and decide which is larger and by how much. 5

20. Look at the pattern: the 6th figure has 50 dots. How many dots are in figure 15? If a figure has 176 dots, which figure is it? Show that the dots form a pattern with common difference 9. 5

OR

Two sequences: A is $3, 10, 17, 24, \ldots$ and B is $40, 43, 46, 49, \ldots$. Find the 15th term of each and decide which is larger and by how much. 5

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND